

# WASP-12b

R-2000

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-12\_150218 · created 2026-08-07T15:56:27+00:00

## BENCHMARK: INACCURATE

### Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2457071.63114 +/- 0.00014 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.1316 +/- 0.0077                 |
| Transit depth (Rp/Rs)^2                           | 1.73 +/- 0.2 [%]                  |
| Orbital Inclination (inc)                         | 76.81 +/- 2.0                     |
| Airmass coefficient 1 (a1)                        | 0.9558 +/- 0.0057                 |
| Airmass coefficient 2 (a2)                        | 0.046 +/- 0.043                   |
| Scatter in the residuals of the lightcurve fit is | 0.6 %                             |
| Best Comparison Star                              | #2 - [537, 252]                   |
| Optimal Aperture                                  | 4.16                              |
| Optimal Annulus                                   | 10.17                             |
| Transit Duration (day)                            | 0.0732 +/- 0.0047                 |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                          |
|--------------------------------------|----------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.1316 $\pm$ 0.0077 vs 0.1178 (deviation 1.72 $\sigma$ ) |
| Duration fit vs published            | 0.0732 d vs 0.125 d (deviation 10.56 $\sigma$ )          |
| Mid-time O-C                         | -0.3 min                                                 |
| Depth SNR                            | 16.4                                                     |
| Residual scatter                     | 0.6 %                                                    |

### Light curve

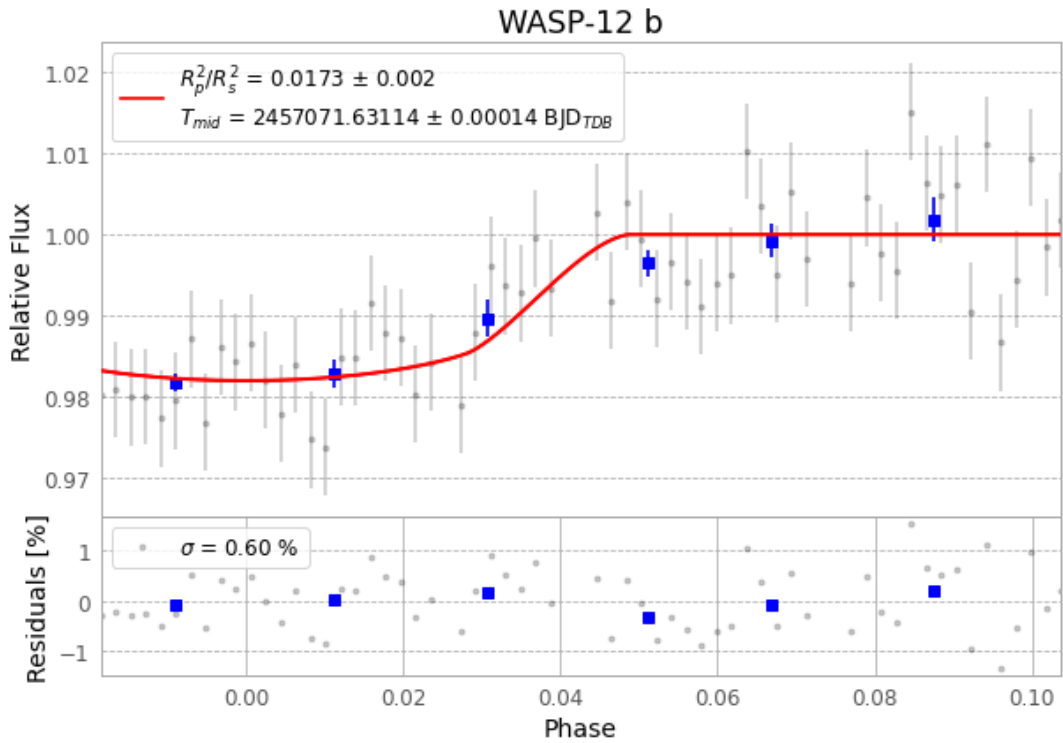


Figure 1 — FinalLightCurve\_WASP-12 b\_2015-02-17.png (SHA-256 607146864e89fb28...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [237, 269], 2 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 0eb0be4dcdcd36e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit fd74a91

## Data provenance

63 FITS frames (41 MB), WASP-12150218023312.FITS → WASP-12150218054512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-150218.log (0115cc696081...), FinalLightCurve\_WASP-12 b\_2015-02-17.png (607146864e89...), FinalLightCurve\_WASP-12 b\_2015-02-17.csv (a6452f3a6fb2...)

## Compute resources

Wall time 86.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 15:56Z.